

O-RAN White Box Hardware Working Group Hardware Reference Design Specification for Indoor Pico Cell with Fronthaul Split Option 6

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Revision History

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Date	Revision	Author	Description
2021.07.08	02.00	WG7	Final version for publication

3

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Chapter 1 Introductory Material

1.1 Scope

This Technical Specification has been produced by the O-RAN ALLIANCE.

In the main body of this specification (in any “chapter”) the information contained therein is informative, unless explicitly described as normative. Information contained in an “Annex” to this specification is always informative unless otherwise marked as normative.

The contents of the present document are subject to continuing work within O-RAN WG7 and may change following formal O-RAN approval. Should the O-RAN.org modify the contents of the present document, it will be re-released by O-RAN ALLIANCE with an identifying change of release date and an increase in version number as follows:

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- y the second digit is incremented when editorial only changes have been incorporated in the document.
- z the third digit included only in working versions of the document indicating incremental changes during the editing process. This variable is for internal WG7 use only.

The present document specifies system requirements and high-level architecture for the Indoor Pico Cell Hardware Reference Design with Fronthaul Split Option 6 deployment scenario as specified in the Deployment Scenarios and Base Station Classes document [1].

1.2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- [1] ORAN-WG7.DSC.0-V01.00 Technical Specification, Deployment Scenarios and Base Station Classes for White Box Hardware
- [2] 3GPP TR 21.905: Vocabulary for 3GPP Specifications
- [3] 3GPP TR 38.104: NR; Base Station (BS) radio transmission and reception
- [4] 3GPP TR 38.913: Study on Scenarios and Requirements for Next Generation Access Technologies
- [5] 3GPP TS 38.401: NG-RAN; Architecture description
- [6] ORAN-WG7.IPC.HAR-v01.00 Technical Specification: O-RAN Indoor Pico Cell Hardware Architecture and Requirement Specification
- [7] SCF 222.10.03: 5G FAPI PHY API Specification
- [8] SCF 082.09.05: nFAPI and FAPI Specifications
- [9] SCF 223.10.02: RF and Digital Front End Control API

[10] SCF 224.10.01: 5G FAPI Network Monitor Mode API

[11] SCF 225.2.0: 5G nFAPI Specifications https://scf.io/en/documents/225_5G_nFAPI_specifications.php

1.3 Definitions and Abbreviations

1.3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [2]. For the base station classes of Pico, Micro and Macro, the definitions are given in 3GPP TR 38.104 [3].

Carrier Frequency: Center frequency of the cell.

F1 interface: The open interface between O-CU and O-DU defined by 3GPP TS 38.473.

Integrated architecture: In the integrated architecture, the O-RU and O-DU are implemented on one platform. Each O-RU and RF front end is associated with one O-DU. They are then aggregated to O-CU and connected by F1 interface.

Split architecture: The O-RU and O-DU are physically separated from one another in this architecture. A switch may aggregate multiple O-RUs to one O-DU. O-DU, switch and O-RUs are connected by the fronthaul interface.

Transmission Reception Point (TRxP): Antenna array with one or more antenna elements available to the network located at a specific geographical location for a specific area.

1.3.2 Abbreviations

For the purposes of the present document, the abbreviations given in [2] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, as in [2].

3GPP	Third Generation Partnership Project
5G	Fifth-Generation Mobile Communications
ADC	Analog to Digital Converter
ASIC	Application Specific Integrated Circuit
BS	Base Station
CFR	Crest Factor Reduction
CISPR	Comité International Spécial des Perturbations Radioélectriques
CU	Centralized Unit as defined by 3GPP
DAC	Digital to Analog Converter
DDC	Digital Down Conversion
DDR	Double Data Rate
DL	Downlink
DPD	Digital Pre-Distortion

1	DPU	Digital Processing Unit
2	DSP	Digital Signal Processor
3	DU	Distributed Unit as defined by 3GPP
4	DUC	Digital Up Conversion
5	ECC	Error-Correcting Code
6	FAPI	Functional Application Platform Interface
7	FCC	Federal Communications Commission
8	FDD	Frequency Division Duplex
9	FFT	Fast Fourier Transform
10	FH	Fronthaul
11	FHGW	Fronthaul Gateway
12	FPGA	Field Programmable Gate Array
13	GbE	Gigabit Ethernet
14	GMC	Grand Master Clock
15	GNSS	Global Navigation Satellite System
16	GPS	Global Positioning System
17	HSUART	High-Speed Universal Asynchronous Receiver/Transmitter
18	IEEE	Institute of Electrical and Electronics Engineers
19	IMC	Integrated Memory Controller
20	I/O	Input/Output
21	JTAG	Joint Test Action Group
22	L2	Layer 2
23	L3	Layer 3
24	LDPC	Low-Density Parity-Check
25	LTE	Long Term Evolution
26	LRDIMM	Load-Reduced Dual In-Line Memory Module
27	MAC	Media Access Control
28	MIMO	Multiple Input Multiple Output
29	MCP	Multi-Chip Package
30	MU-MIMO	Multiple User MIMO
31	nFAPI	network Functional Application Platform Interface
32	NFV	Network Function Virtualization
33	NMEA	National Marine Electronics Association
34	NPU	Network Processing Unit
35	NR	New Radio
36	O-CU	O-RAN Centralized Unit as defined by O-RAN
37	O-DU ₆	A specific O-RAN Distributed Unit having fronthaul split option 6
38	O-RU ₆	A specific O-RAN Radio Unit having fronthaul split option 6
39	PCIe	Peripheral Component Interface Express
40	PCH	Platform Controller Hub
41	PDCP	Packet Data Convergence Protocol
42	PHY	Physical layer
43	PLL	Phase Locked Loop
44	PMBus	Power Management Bus

1	PoE	Power over Ethernet
2	PWM	Pulse Width Modulation
3	RAN	Radio Access Network
4	RDIMM	Registered Dual In-Line Memory Module
5	RFFE	Radio Frequency Front End
6	RGMII	Reduced Gigabit Media-Independent Interface
7	RF	Radio Frequency
8	RFFE	RF Front End
9	RoHS	Restriction of Hazardous Substances
10	RU	Radio Unit as defined by 3GPP
11	RX	Receiver
12	SFP	Small Form-factor Pluggable
13	SFP+	Small Form-factor Pluggable Transceiver
14	SOC	System On Chip
15	SPI	Serial Peripheral Interface
16	TDP	Thermal Design Power
17	TR	Technical Report
18	TS	Technical Specification
19	TX	Transmitter
20	UART	Universal Asynchronous Receiver/Transmitter
21	UL	Uplink
22	USB	Universal Serial Bus
23	WG	Working Group
24		
25		

Chapter 2 Hardware Reference Design

2.1 O-CU Hardware Reference Design

The O-CU white box is the hardware platform that performs the CU function of upper L2 and L3. The hardware systems specified in this document meet the computing, power and environmental requirements for O-RAN deployment scenarios. These requirements are described in the hardware requirement specification as well as in the deployment scenarios. The O-CU hardware includes the chassis platform, mother board, peripheral devices and cooling devices. The mother board contains processing unit, memory, the internal I/O interfaces, and external connection ports. The midhaul and backhaul interfaces are used to carry the traffic between O-CU and O-DU₆ as well as O-CU and core network. Other hardware functional components include storage for software, debugging interfaces, and board management controller just to name a few. The O-CU designer will make decisions based on the specific needs of the implementation.

As discussed in the hardware architecture requirements [6], the O-CU white box hardware is similar to O-DU₆ on interfaces and processing unit. Rather than specifying a dedicated O-CU reference design, we reuse O-DU₆ hardware reference design for O-CU.

2.2 O-DU Hardware Reference Design

The terms “O-DU₆” and “O-RU₆” are used here for O-DU and O-RU that use split option 6. O-DU₆ hardware systems specified in this document meet the computing, power and environmental requirements of the indoor picocell deployment scenario configuration. The O-DU₆ hardware includes the chassis platform, mother board, peripheral devices and cooling devices. The mother board contains processing unit, memory, the internal I/O interfaces, and external connection ports. The front haul and back haul interface are used to carry the traffic between O-DU₆ and O-RU₆ as well as O-CU and O-DU₆.

The O-DU₆ white box is the hardware platform that performs MAC functions while PHY is done in the O-RU₆. The O-DU₆ design may also provide an interface for hardware accelerator(s) if that option is preferred. However, this design does not specify accelerator hardware. Other hardware functional components include storage for software, debugging interfaces, and board management controller. The O-DU₆ designer will make decisions based on the specific needs of the implementation.

Note that the O-DU₆ HW reference design is also reusable for O-CU and integrated O-CU/ O-DU₆.

2.2.1 O-DU High-Level Functional Block Diagram

O-DU₆ and O-RU₆ are connected via nFAPI interfaces over fiber/ethernet transport. With an option 6 MAC/PHY fronthaul split, the O-RU₆ includes complete physical layer processing functions and

the O-DU₆ handles higher layer processing functions. The O-DU₆ hardware architecture is shown in **Error! Reference source not found..** Note that O-DU₆ does not require an accelerator module since physical layer functions are handled by O-RU₆.

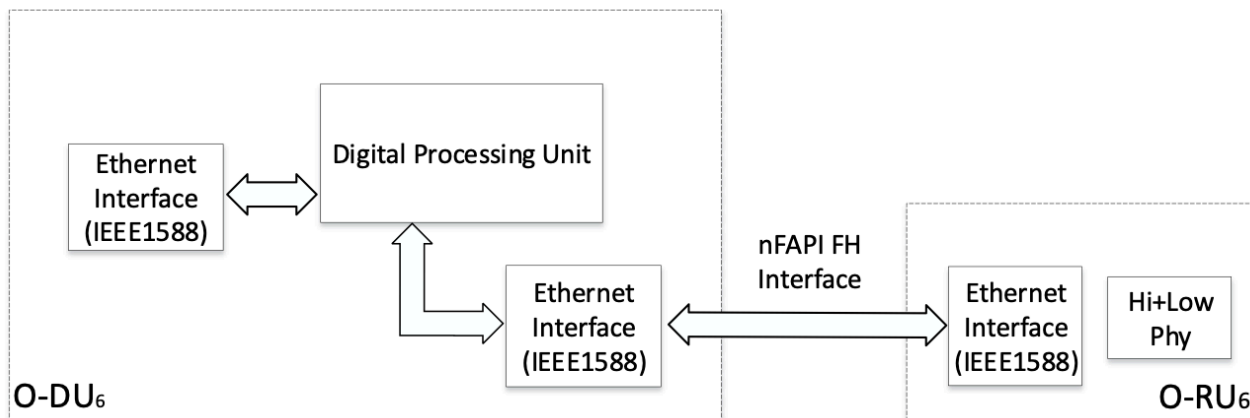


Figure 2-1 O-DU₆ Architecture Diagram

2.2.2 O-DU₆ Hardware Components

The sections below describe the functionality, interface and performance for each respective block of the O-DU₆. As device integration is an ongoing activity, chip boundaries may be fluid and some functionalities may move from one block to another or entire functionalities may be absorbed into other blocks. The sections below describe the functional blocks independent of which physical device they may reside in.

2.2.2.1 Digital Processing Unit

a. Digital Processing Unit Requirement

General purpose processor performance and other related information are listed in Table 2-1.

Table 2-1: Processor Feature List

Item Name	Description
# of Cores	16
# of Threads	32
Base Frequency	2.20 GHZ
Max Turbo Frequency	3.00 GHz
Cache	22 MB
TDP	100W
Max Memory Size (dependent on memory type)	512 GB
Memory Types	DDR4
Max # of Memory Channels	4

Interfaces

The interface specification in main board are as follows:

➤ Memory Channel Interfaces

The system memory capacity, type and related information are described in Table 2-2.

Table 2-2: Memory Channel Feature List

Item Name	Description
Memory Types	DDR4
# of Memory Channels	4
ECC LRDIMM	Up to 512GB
ECC RDIMM	Up to 256GB
Memory Speed	2666/2400/2133MHz
DIMM Sizes	128GB, 64GB, 32GB, 16GB
Memory Voltage	1.2 V

➤ PCIe

PCIe Gen 3 is supported by the processor. There are a total of 32 PCIe lanes with 128 Gb/s bandwidth. The 32 PCIe lanes can be divided into two x16 slots by using a riser card.

➤ Ethernet

The system should be capable to offer aggregated 48 Gb/s Ethernet bandwidth. When higher Ethernet bandwidth is required, an Ethernet card can be installed in one of the PCIe slots.

b. Digital Processing Unit Design

The digital processing unit is a system-on-a-chip (SOC) device which is a 64-bit, multi-core server class processor. This SoC includes an integrated Platform Controller Hub (PCH), integrated high-speed I/O, Integrated Memory Controllers (IMC), and four integrated 10 Gigabit Ethernet ports. The SOC supports 512-bit wide vector processing instruction set. It also supports hardware virtualization to enable dynamic provisioning of services as communication service providers extend network function virtualization (NFV). **Error! Reference source not found.** shows the major functional blocks of the digital processing unit.

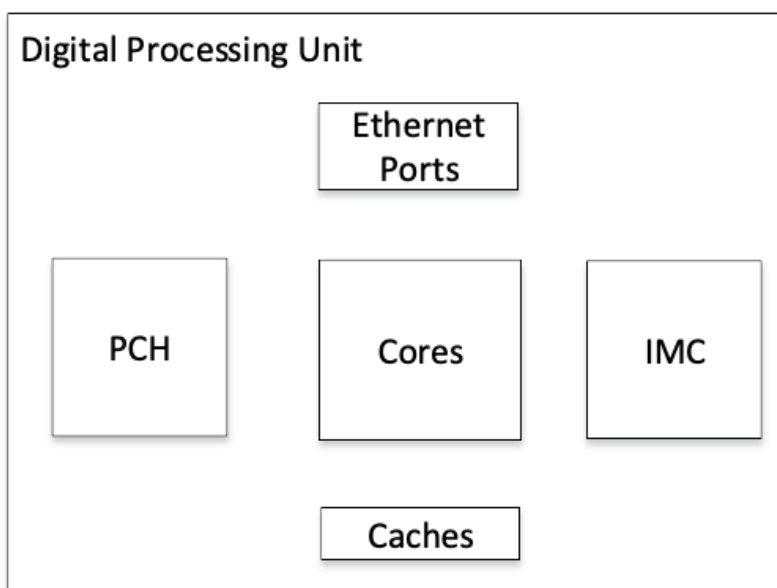


Figure 2-2 Digital Processing Unit Block Diagram

2.2.2.2 Hardware Accelerator

Hardware accelerators may be used in O-DU₆ to offload computationally intensive functions and to optimize the performance under varying traffic and loading conditions. The acceleration functional requirements and implementation are system designer's choice; however, the O-DU₆ shall meet the minimum system performance requirements under various loading and deployment conditions. In most cases, an FPGA or ASIC based PCIe card can be used to optimize the system performance. Examples for O-DU₆ could be security, network interface controller, and general packet acceleration. Specific accelerator designs are not called out in this specification.

2.2.3 Synchronization and Timing

2.2.3.1 Synchronization and Timing Design 1

The timing synchronization of O-DU₆ is achieved by utilizing IEEE1588 via Ethernet ports. The Network Interface Controllers (NIC) that are used for fronthaul and backhaul support the IEEE1588 protocol.

a. Hardware Requirements

The O-DU₆ shall support at least one of following timing synchronization methods:

1. Ethernet based IEEE1588
2. GNSS based timing synchronization.

b. Hardware Design

Depending on the timing distribution topologies used, the O-DU₆ system clock is able to synchronize with the Grand Master Clock (GMC) using IEEE1588 via either the fronthaul NIC or backhaul NIC, or synchronizing timing using Global Navigation Satellite System (GNSS). In the case of IEEE1588, the Physical Hardware Clock (PHC) within the NIC is synchronized with the GMC first, then the O-DU₆ system clock is synchronized with the PHC. The O-DU₆ is also capable to provide clock to the O-RU₆ via fronthaul if needed. When GNSS is available to O-DU₆, it is able to synchronize the system clock to Coordinated Universal Time (UTC). Figure 2-3 outlines the O-DU₆ timing synchronization mechanisms.

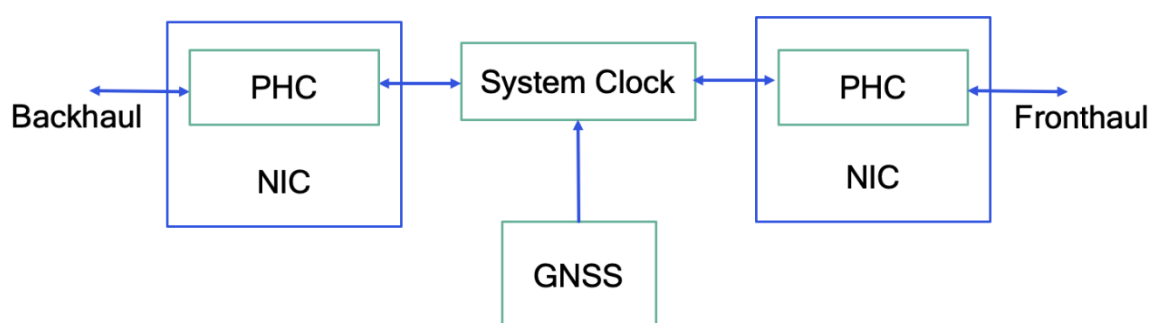


Figure 2-3 O-DU₆ Timing Synchronization

2.2.4 External Interface Ports

a. Hardware Requirements

Table 2-3 External Port List 2-3 shows the external ports or slots that the system provides.

Table 2-3 External Port List

Port Name	Feature Description	19
Ethernet	Octave Gigabit Ethernet LAN connectors	
	Dual 10GbE Base-T Ethernet connectors	
	Dual 10GbE SFP + Fiber Ethernet connectors	
USB	2 USB 3.0 ports	
Serial Port	1 COM port via RJ45	
Antenna port	1 SMA connector for GNSS	

b. Hardware Design

The digital processing unit is a SOC device which provides all the external ports described in the hardware requirement section. The system includes 2 USB 3.0 ports, and the serial RS232 port that

can be used for Console Redirection, e.g. Out-of-Band Management/Windows Emergency Management Services (EMS). There are a total of four 10G Ethernet ports in system, among them are two RJ45 10GbE and two 10G SFP+ ports. The system also provides a RF interface to connect GNSS antenna.

2.2.5 Mechanical

➤ Mother Board

The mechanical layout of the mother board in Figure 2-4 shows the location of major components and interface ports, and the dimension of the board.

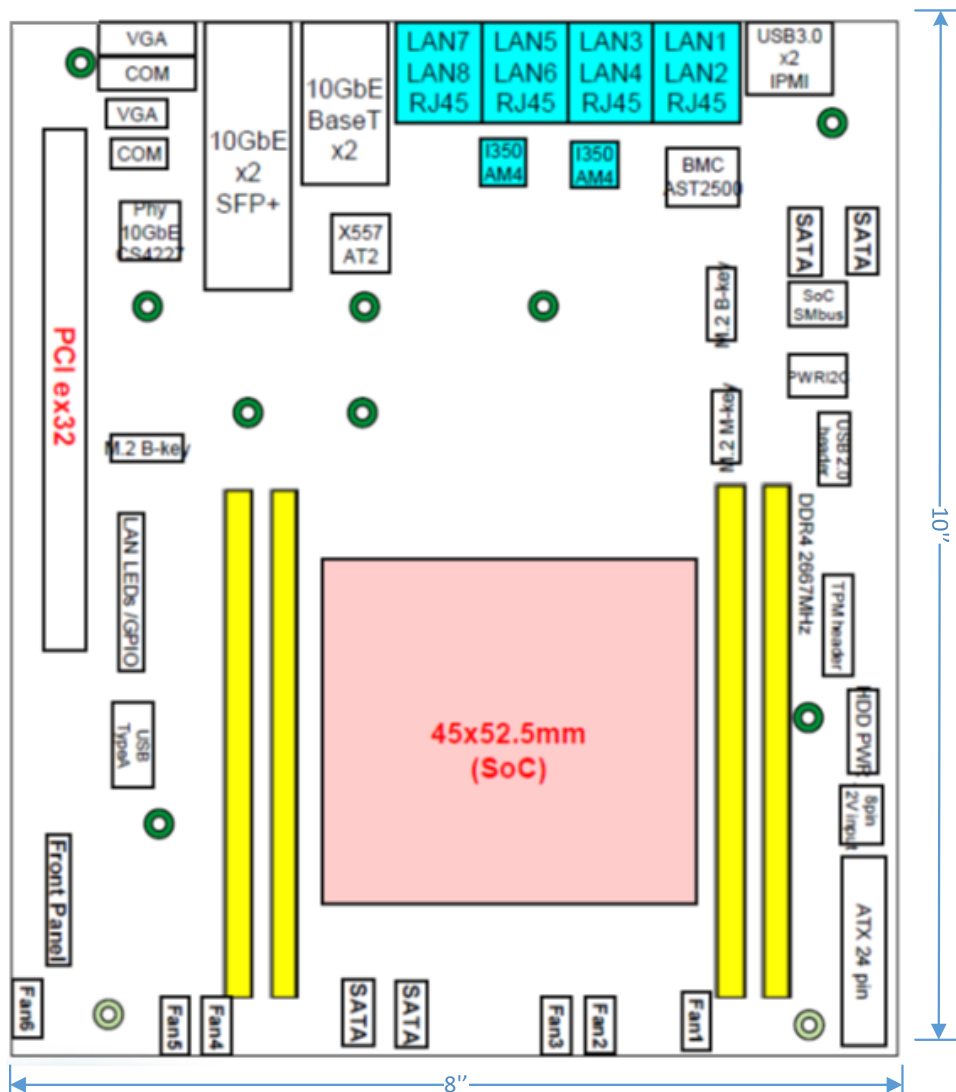


Figure 2-4 Mother Board Layout

➤ Chassis

The 1U rack mount chassis contains the layout of the power supply, SSD and fans. The chassis dimension is shown in Figure 2-5.

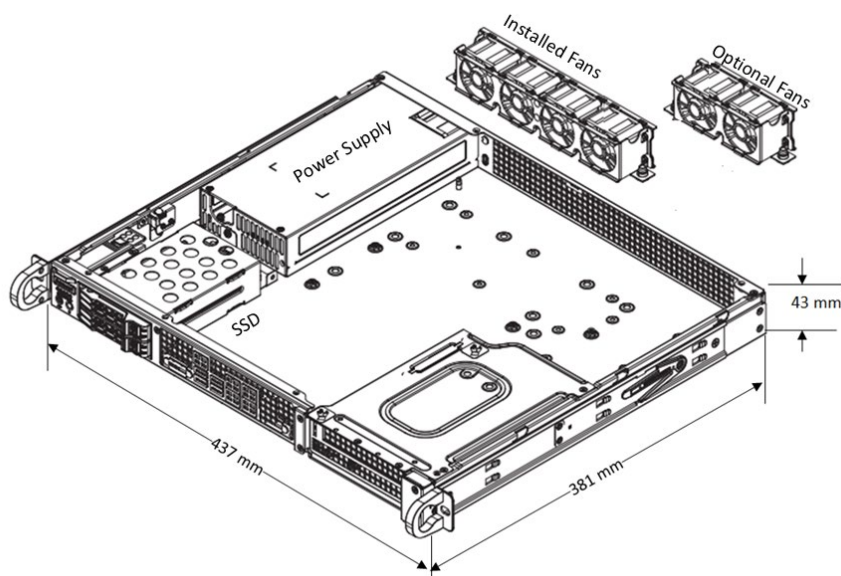


Figure 2-5 Chassis Mechanical Diagram

➤ Cooling

The system has 4x 40x28mm PWM fans for cooling. Up to 6 fans can be installed if needed.

2.2.6 Power Unit

In a fully loaded system with two PCIe slots populated with 75W each, the system power consumption should be less than 400W. The total system power shall be kept to less than 80% of the power supply capacity.

a. Hardware Requirements

The power is provided by 500W High-Efficiency power supply with PMBus 1.2. The power support input and output power rails are listed below.

AC Input: 100-240V, 50-60Hz, 6.6A max

DC Output: +3.3V: 12A

+5V: 15A

+5V standby: 3A

+12V: 41A

-12V: 0.2A

b. Hardware Design

The O-DU₆ chassis includes one 500W power supply unit. The power supply unit is auto-switching capable, which enables it to automatically sense and operate at a 100v to 240v input voltage. The power supply unit features are listed in Table 2-4.

Table 2-4 Power Supply Features

Features	Parameters
Output connectors	24pin/8pin/4+4pin/HDD/I2C
Dimension (W x D x H)	3.9 x 7.1 x 1.6 inch
Maximum Output Power	+3.3V: 12A +5V: 15A +12V: 41A -12V: 0.2A +5Vsb: 3A
Rated Input Voltage/Current	100-240Vac / 6.6A max
Rated Input Frequency	50-60HZ
Inrush current	Less than 30A

2.2.7 Thermal

Active cooling with up to 6 fans integrated in chassis.

2.2.8 Environmental and Regulations

The O-DU₆ hardware system is RoHS Compliant. The power supply unit is EMC FCC/CISPR Class B compliant. Table 2-5 lists the environmental related features and parameters.

Table 2-5 Environmental Features

Features	Description
Operating Temperature	-5°C to 55°C
Non-operating Temperature	-40°C to 70°C (-40°F to 158°F)
Operating Relative Humidity	8% to 90% (non-condensing)
Non-operating Relative Humidity	5% to 95% (non-condensing)

2.3 O-RU₆ Hardware Reference Design

This section provides a hardware reference design for O-RU₆, that is an O-RU built using the MAC/PHY split and interfaces defined in the Small Cell Forum. See references [7], [8], [9], [10] and [11].

2.3.1 O-RU₆ High-Level Functional Block Diagram

Figure 2-6 shows a high-level functional block diagram depicting the major HW/SW components and internal/external interfaces for O-RU₆. In this design, PHY processing in the O-RU₆ is handled by a specialized ASIC. O-RU₆ can handle fiber or metallic ethernet transport interface ports.

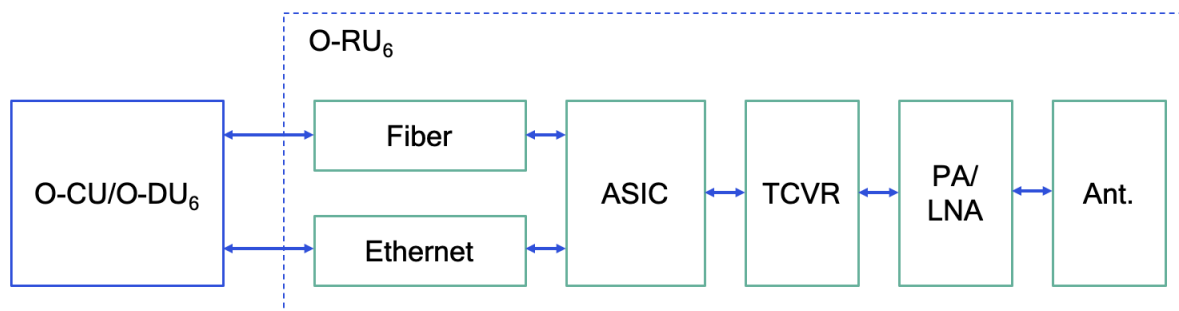


Figure 2-6 General Block Diagram for O-RU₆

2.3.2 O-RU₆ Hardware Components

The generic hardware architecture for O-RU₆ consists of three major parts: digital processing unit, RF processing unit and a timing unit as shown in Figure 2-7. In the next section there is a brief mapping of modules within these major blocks of O-RU₆.

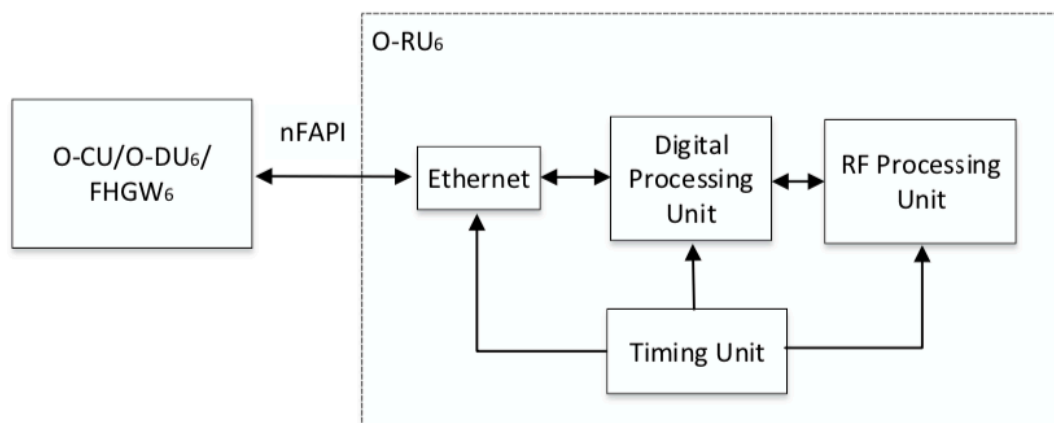


Figure 2-7 O-RU₆ Architecture Diagram

Dissimilar to option 7-2 or option 8, all L1 processing is performed by O-RU₆. The processing unit handles transport and interface processing as well as all L1 modem processing. The DPU in O-RU₆ is composed of an NPU to terminate transport layer and a PHY ASIC for L1 processing. Figure 2-8 **Error! Reference source not found.** shows the major components used in O-RU₆.

The digital processing unit handles transport/interface processing, complete PHY (low and high PHY functions) modem processing, Digital Down Converter (DDC), Digital Up Converter (DUC), Crest Factor Reduction (CFR) and Digital Pre-Distortion (DPD).

The RF processing unit is comprised of the transceiver block which is responsible for functions such as Analog to Digital Converter (ADC), Digital to Analog Converter (DAC) and mixer; an amplification block which contains Power Amplifier (PA)/ Low Noise Amplifier (LNA), etc; and antenna subsystem that captures and/or transmits radio electromagnetic waves. O-RU₆ may use Power over Ethernet (PoE) to simplify deployment complexity.

The Timing Unit may include Phase Locked Loop (PLL), local oscillator and timing synchronization circuitry.

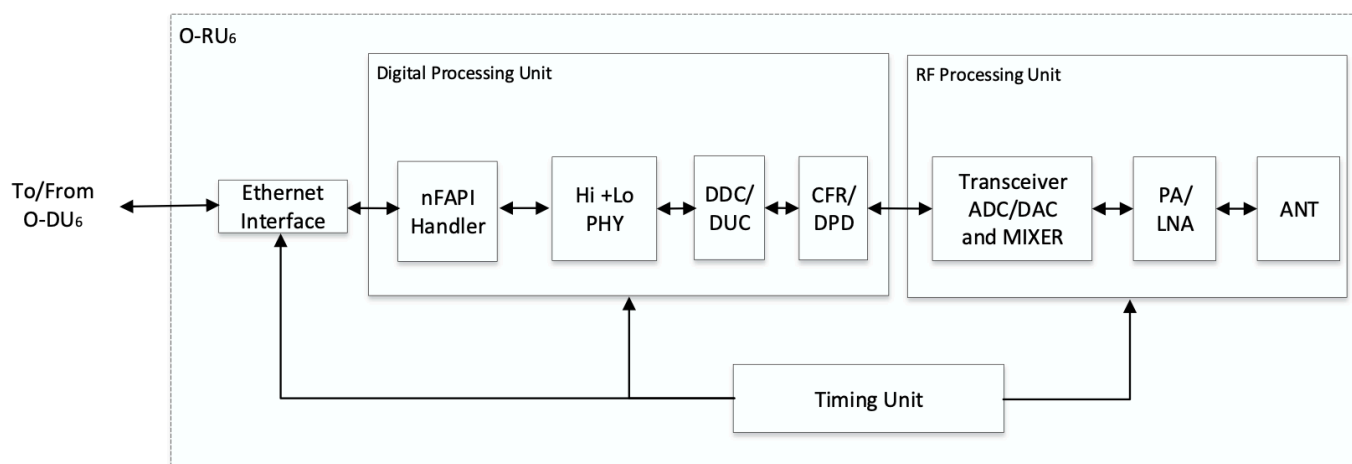


Figure 2-8 O-RU₆ Functional Module Diagram

2.3.3 O-RU₆ Functional Module Description

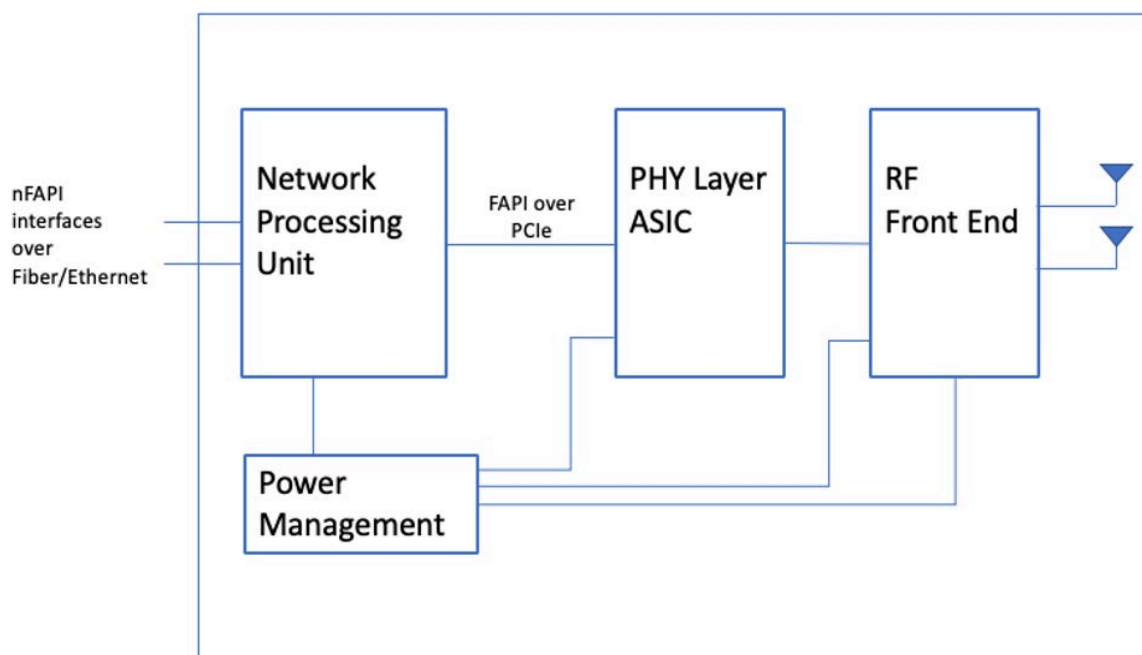


Figure 2-9 Sub-6 GHz Indoor Pico O-RU₆ Functional Modules (Split Option 6)

Mapping the generic hardware architecture to this hardware reference design, the DPU is split to an NPU and PHY ASIC, with an internal interface using FAPI over a PCIe interface.

Functional blocks shown in Figure 2-9 are discussed below.

2.3.3.1 Network Processing Unit (NPU)

The Network Processing Unit (NPU) for O-RU₆ terminates the external nFAPI interfaces and provides the internal FAPI interface to the PHY ASIC.

2.3.3.1.1 Networking I/O

- Minimum I/O is a single 10GbE port that can be used with electrical or optical I/O
- Optionally, support can be added for
 - OAM / maintenance port (1GbE) connecting over RGMII to the NPU
 - An additional 10GbE port for chaining or failover connectivity. By implication, the NPU needs to support efficient local switch/routing functionality

Networking side connectivity is shown as below:

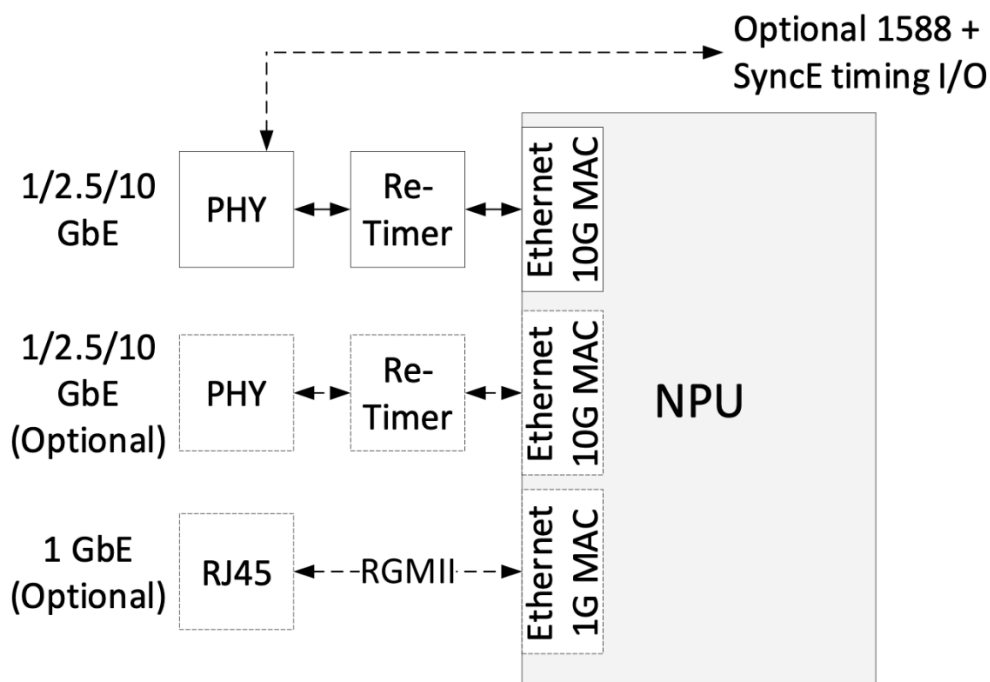


Figure 2-10 O-RU₆ Networking I/O

Example SFP connectivity is shown below:

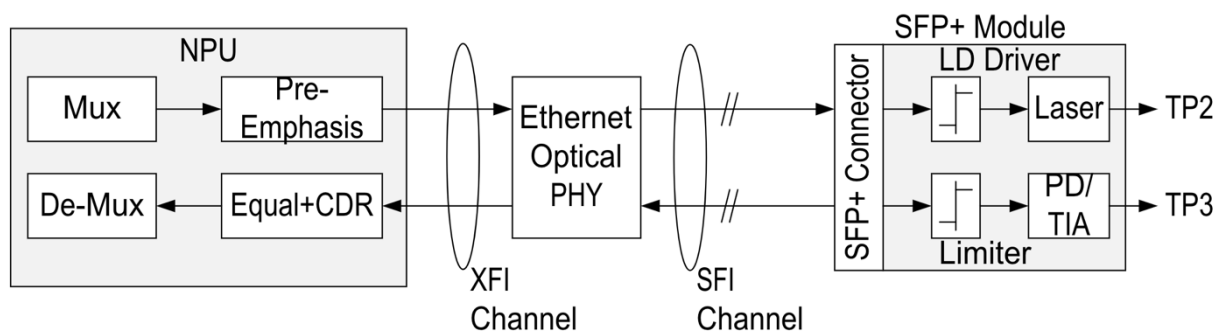


Figure 2-11 O-RU₆ SFP Connectivity

2.3.3.1.2 Low-speed connectivity and boot

This picture shows typical low-speed I/O associated with the NPU

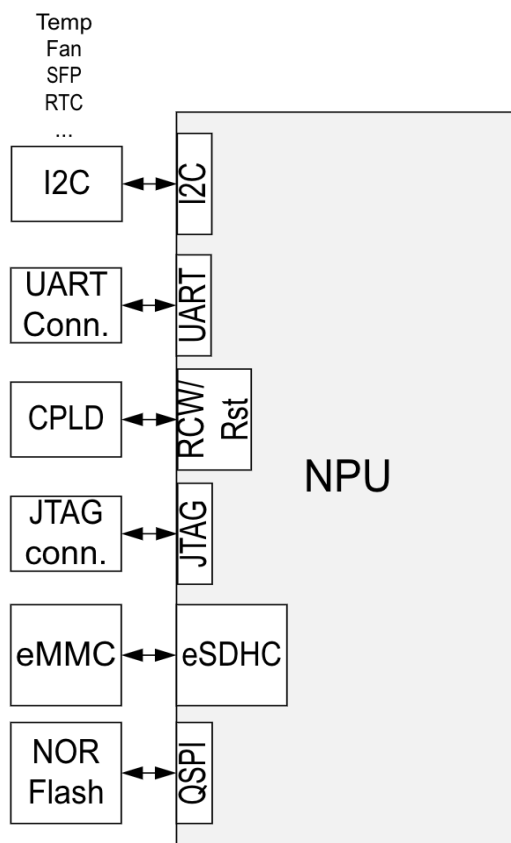


Figure 2-12 O-RU₆ Low-speed NPU Connectivity

In addition, the device can optionally be configured for secure boot.

2.3.3.1.3 DDR

- Assuming a telecommunications-tuned Linux image and virtualization through containerization, accommodate for 512Mbyte for hosting the Linux OS from DDR
- Buffering needs to be provided for FAPI messages, but this is expected to be limited to few Mbyte (marginal) as well as debug / OAM logging functionality
- In aggregate, assume ~1Gbyte aggregate DDR dimensioning
- For bandwidth estimation, assume ~5Gbps of Ethernet throughput with 2 paths to/from the DDR interface

- Receive / Transmit from / to Ethernet MAC to DDR
- Receive / Transmit from / to DDR to PCIe

Aggregate is ~10Gbps or 15Gbps with margin. Given a maximum utilization of the DDR interface of 50%, we require 30Gbps of raw memory throughput. A 1.6GTPS DDR memory controller with 32bits memory bus width provides 50Gbps throughput and is thus enough to support this use-case as well as additional bandwidth requirements for switch/routing functions to optional Ethernet ports.

- ECC is assumed to be a required for a telecom-grade application.

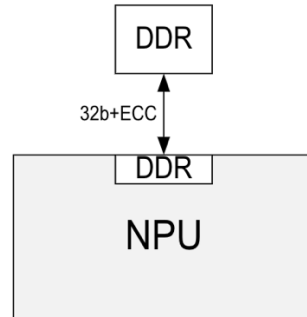


Figure 2-13 O-RU₆ DDR

2.3.3.1.4 Processor subsystem

- The NPU needs to incorporate a processor subsystem that is capable of processing networked FAPI messages and convert them to PCI messages. Dimensioning of the CPU is outside of the scope.

2.3.3.1.5 Modem I/O

- For bandwidth estimation, assume ~5Gbps of throughput. Adding 20% of signaling overhead, this becomes ~6Gbps.
- Assuming PCIe gen3 operation, 2 lanes of PCIe @8Gbps provide sufficient overhead above this bandwidth.

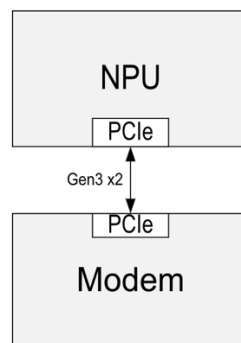


Figure 2-14 O-RU₆ Modem I/O

2.3.3.1.6 Security

- The NPU needs to support high-speed ciphering and protocol processing capabilities to support a (IPSec) secured backhaul interface. This can be achieved through integrated networking acceleration or an external accelerator as shown below.

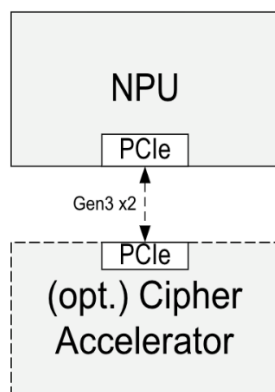


Figure 2-15 O-RU₆ NPU Security

2.3.3.2 PHY Layer ASIC

The PHY Layer ASIC is an application specific integrated circuit that provides a highly integrated and power-efficient solution implementing the complete 5G-NR PHY layer. It includes variants for both sub-6 GHz (FR1) and mmWave (FR2) frequency ranges. These ASICs can be used to implement self-contained, all-in-one small cells, as well as Option 6 radio units (RU) for disaggregated RAN systems. In either case, the ASIC connects to a co-located network processing unit (NPU) via a PCIe interface. In the case of a self-contained small cell, the NPU hosts the full L2/L3 protocol stack (i.e., RLC, MAC, PDCP) which connects to the PHY via the 5G FAPI logical interface over PCIe. In the case of an Option 6 RU, the NPU terminates the 5G nFAPI fronthaul connection between the CU/DU and the RU and connects to the PHY via the aforementioned 5G FAPI logical interface (over PCIe.)

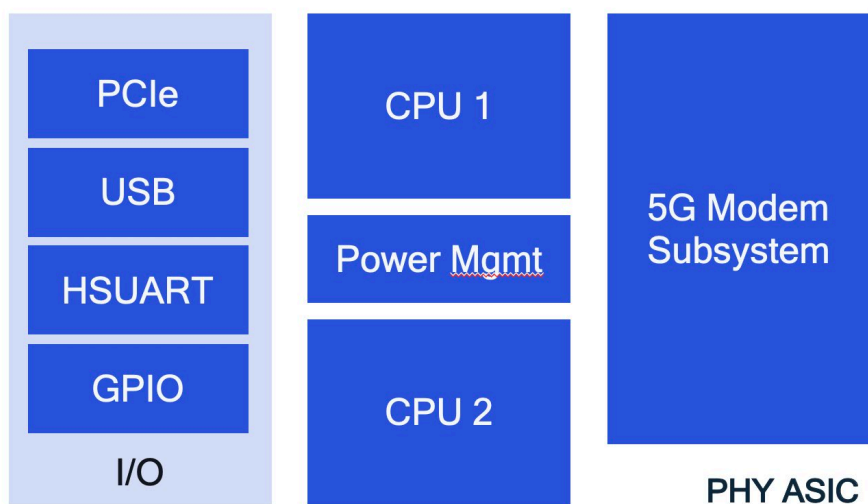


Figure 2-16 O-RU₆ PHY ASIC block diagram

Figure above provides a high-level diagram of the functional blocks that make up the PHY ASIC. External interfaces, including PCIe, USB, HS-UART, and GPIOs, are shown on the left-hand side. There are two embedded general-purpose CPUs available as well as a power management subsystem. The most important block is the 5G-NR Modem Subsystem.

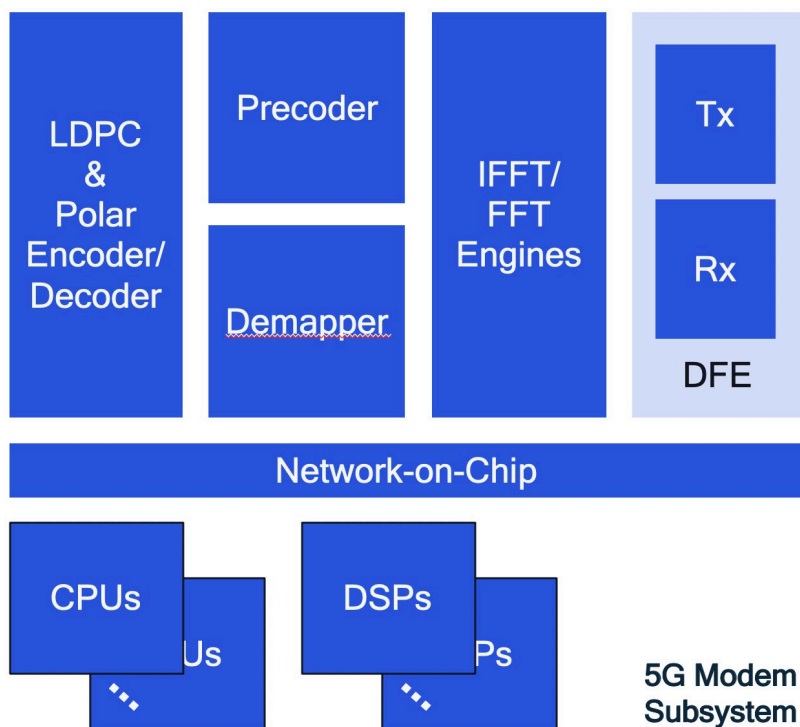


Figure 2-17 O-RU₆ 5G Modem Subsystem

Figure above depicts a block diagram of the Modem Subsystem within the ASIC, which is predominantly software defined, leveraging a number of embedded DSPs and CPUs which together implement the transmit and receive functionalities of the 5G modem. The modem also includes hardware accelerators for computationally intensive tasks such as FFTs, demapping, and LDPC encoding/decoding. The various functional blocks are interconnected via a network-on-chip (NOC). The modem subsystem also includes a digital front-end responsible for signal conditioning in both the transmit (downlink) and receive (uplink) directions.

2.3.3.3 RF Front End

The Radio Frequency Front End (RFFE) is made up of all components between the modem and antennas. These include RF transceivers, RF switches, filters, duplexers, power amplifiers, and low-noise amplifiers.

2.3.3.4 Power Management

The Power Management functional module controls the flow of electrical power to other parts of the O-RU₆.

2.3.4 Synchronization and Timing

2.3.4.1 Hardware Requirements

O-RU timing synchronization sources include:

- Ethernet based IEEE1588
- Ethernet based Synchronous Ethernet
- Optional Global Navigation Satellite System (GNSS) based timing synchronization

Timing functionality is shown in the diagram below:

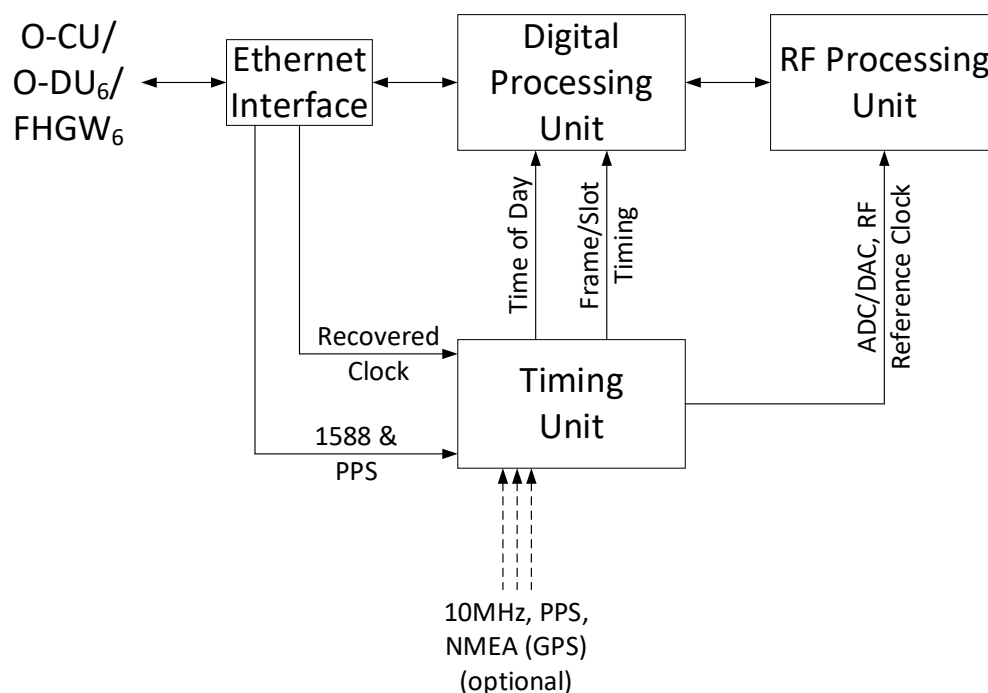


Figure 2-18 Timing Modules

IEEE1588 time stamping can be done at the Ethernet PHY level (model 1 below) or at the Ethernet MAC/Network Processing Unit (NPU) level (model 2 below). Model 1 is expected to provide incremental precision.

2.3.4.2 Synchronization and Timing Design 1

Timing synchronization on Ethernet PHY based 1588 and Synchronous Ethernet received clock extraction support.

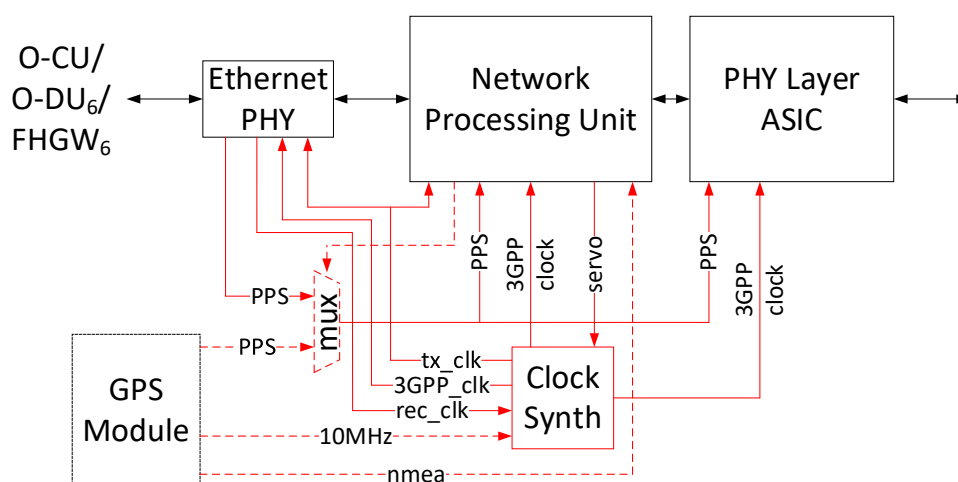


Figure 2-19 Ethernet Synchronization I

Notes:

- Either (depending on mux / input control) GPS 10MHz or Synchronous Ethernet recovered output clock (ref_clk) are input to a clock synthesizer (fractional PLL) to generate the 38.4, 122.88 MHz (or similar) MHz reference clock (Frequency Synchronization)
- The PPS pulse from the GPS provides a mechanism to manage local 122.88 (or similar) 3GPP clock through use of a Software PLL (servo) in absence of GPS/Synchronous Ethernet extracted reference frequency
- On top of the GPS PPS pulse, the GPS UART output provides time synchronization to a universal clock through NMEA messages
- The mux in the system allows the Network Processing Unit to control which input is used as PPS source (GPS module or Ethernet PHY)
- All dashed lines are indicating optionality – GPS support is optional for this system
- IEEE1588 provides both time synchronization and frequency synchronization of a local 122.88MHz clock through Ethernet time stamping such as supported in Layerscape devices. Synchronous Ethernet can be used in conjunction with this for local 3GPP clock generation.

2.3.4.3 Synchronization and Timing Design 2

Timing synchronization on Network Processing Unit based 1588 and Synchronous Ethernet received clock extraction support.

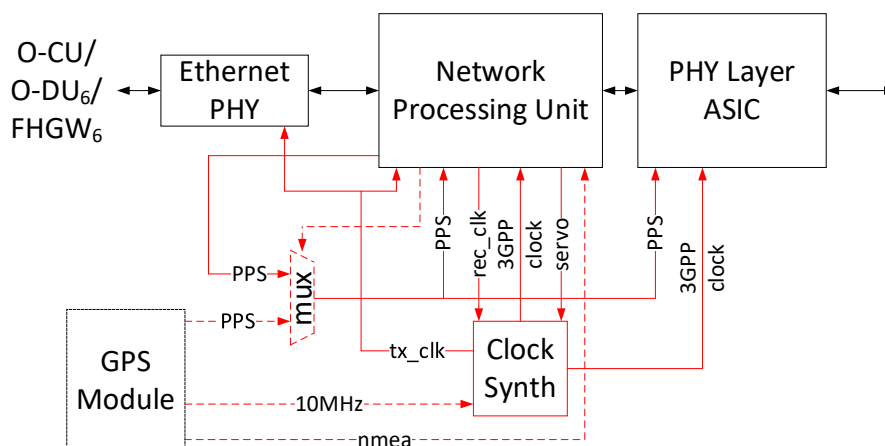


Figure 2-20 Ethernet Synchronization II

Notes:

- Either (depending on mux / input control) GPS 10MHz or Synchronous Ethernet recovered output clock (ref_clk) are input to a clock synthesizer (fractional PLL) to generate the 38.4, 122.88 MHz (or similar) MHz reference clock (Frequency Synchronization)
- The PPS pulse from the GPS provides a mechanism to manage local 122.88 (or similar) 3GPP clock through use of a Software PLL (servo) in absence of GPS/Synchronous Ethernet extracted reference frequency
- On top of the GPS PPS pulse, the GPS UART output provides time synchronization to a universal clock through NMEA messages
- All dashed lines are indicating optionality – GPS support is optional for this system
- IEEE1588 provides both time synchronization and frequency synchronization of a local 122.88MHz clock through Ethernet time stamping such as supported in Layerscape devices. Synchronous Ethernet can be used in conjunction with this for local 3GPP clock generation.

2.3.5 External Interface Ports

The O-RU₆ must have at least one RJ45 10 gigabit per second (Gbps) Ethernet interface or at least one 10 Gbps optical interface. In some cases, depending on the fronthaul throughput associated with smaller air interface bandwidths, a 1 or 2.5 Gbps Ethernet or optical interface may be sufficient, depending on system operator requirements. SCF 5G nFAPI should be supported. At full load, the power consumption must not exceed the limit allowed by IEEE802.3at/PoE+, which is 25.5W.

2.3.6 Power Unit

O-RU₆ can use power over Ethernet (PoE) to simplify deployment complexity.

2.3.7 Environmental and Regulations

O-RU₆ IPC design is for indoor environments (nominal ambient temperature, humidity, altitude). Suppliers must comply with any local regulatory requirements.

2.4 Integrated gNB-DU Reference Design Solution

This Section includes a reference design for an all-in-one, integrated gNB-DU solution. This section provides an example white box design for integrated indoor picocells, using FAPI as the internal interface between PHY and MAC. Typically, the MAC will run on an NPU while the PHY resides in a PHY Layer ASIC, connected via a PCIe interface.

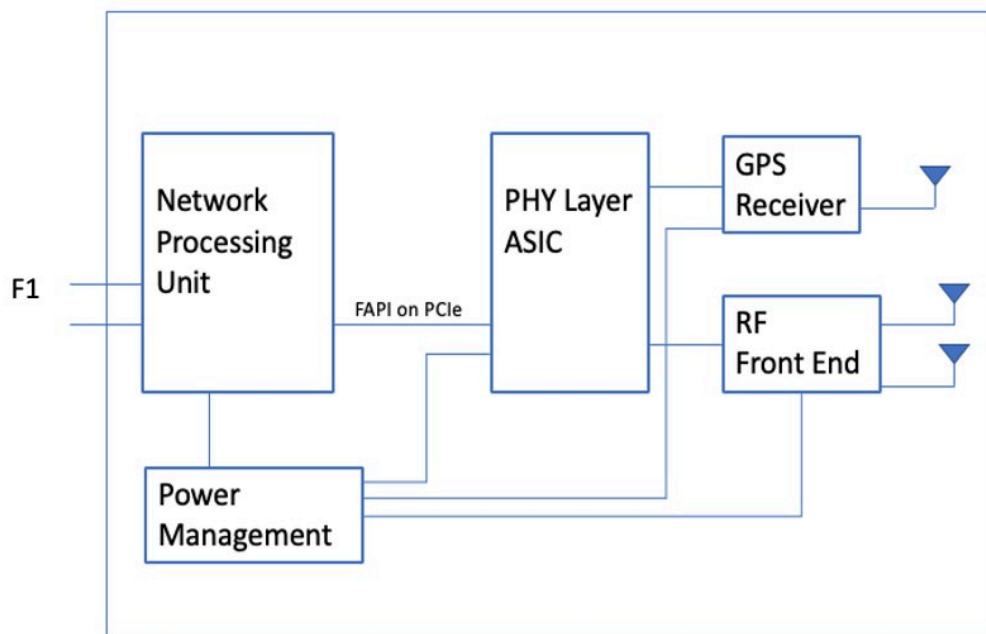


Figure 2-21 Sub-6 GHz Indoor Pico Integrated O-DU&O-RU Functional Modules

Functional modules shown in Figure 2-21 for the integrated case are essentially the same as those in the O-RU₆ split option. The main difference is in the NPU, which now terminates the O-CU/gNB-DU interface.

The integrated gNB-DU may also include a GPS receiver to locally provide location and precise timing (as opposed to depending on the O-CU for those functions.)

The integrated Option 6 design does not require an accelerator.

As in the split design, the DPU is a combination of NPU and PHY ASIC (same parts whether split or integrated.)

2.4.1 Power Unit

If PoE is used, At full load, the power consumption must not exceed the limit allowed by IEEE802.3at/PoE+, which is 25.5W.

2.4.1.1 Power Management

The Power Management functional module controls the flow of electrical power to other parts of the integrated O-DU/O-RU.

2.4.2 Synchronization and Timing

Local 1588v2 or GPS Slave interfaces for acquiring timing and synchronization can be included according to operator use case and deployment environment.

2.4.3 External Interface Ports

The integrated O-DU/O-RU must have at least one RJ45 10 gigabit per second (Gbps) Ethernet interface or at least one 10 Gbps optical interface. In some cases, depending on the fronthaul throughput associated with smaller air interface bandwidths, a 1 or 2.5 Gbps Ethernet or optical interface may be sufficient, depending on system operator requirements. SCF 5G nFAPI should be supported.

2.4.4 Power Unit

Integrated O-DU/O-RU can use power over Ethernet (PoE) to simplify deployment complexity. At full load, the power consumption must not exceed the limit allowed by IEEE802.3at/PoE+, which is 25.5W.

2.4.5 Environmental and Regulations

Integrated O-DU/O-RU IPC design is for indoor environments (nominal ambient temperature, humidity, altitude). Suppliers must comply with any local regulatory requirements.

2.5 FHGW Hardware Reference Design

A FHGW is not required for option 6 hardware reference design.

Annex 1: Example Hardware Bill of Materials

This annex shows example components used to build the reference design described above for split and integrated option 6 designs (note that the same components are used for either scenario.)

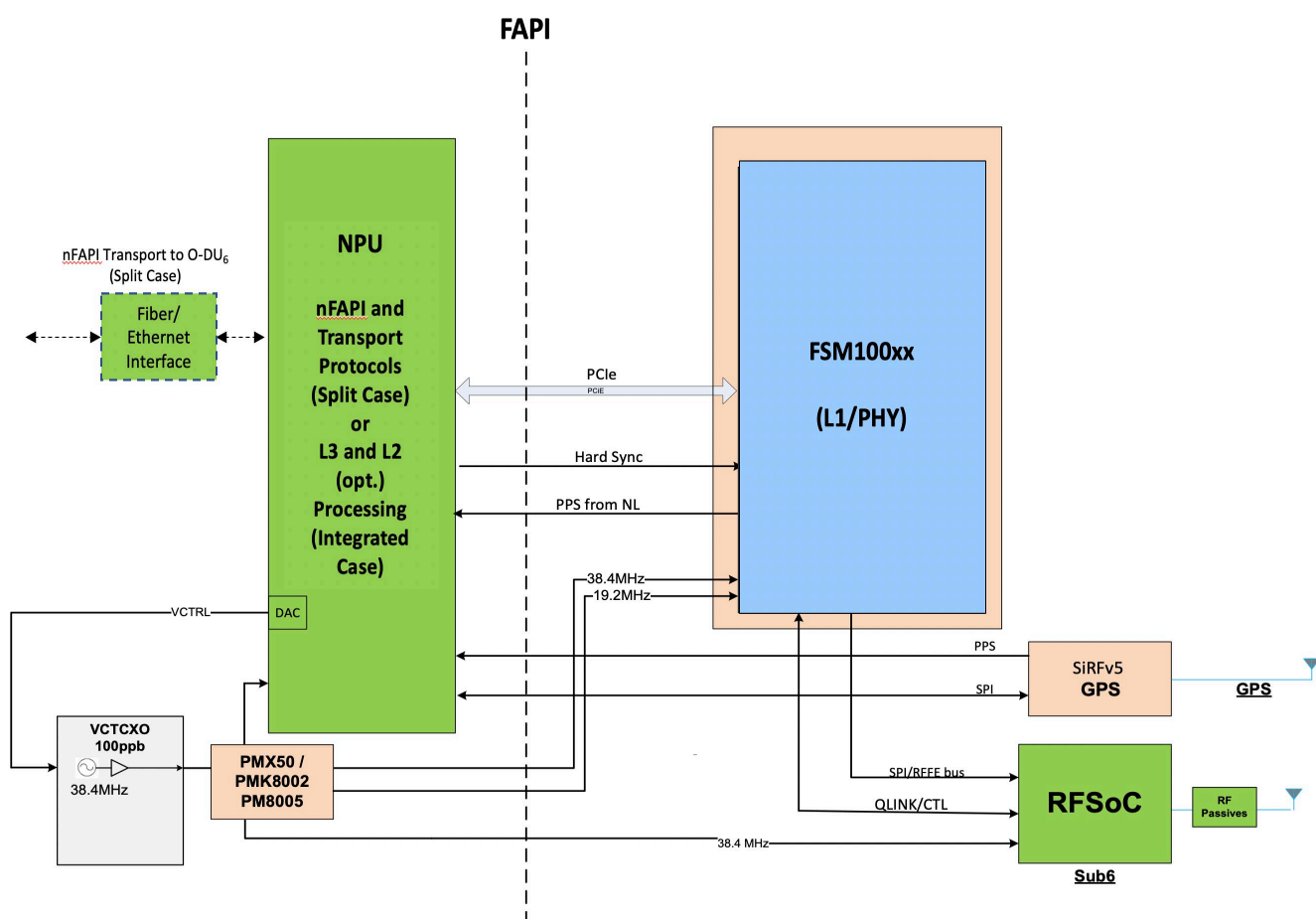


Figure A-1: Sub-6 Split (O-RU₆) and Integrated Picocell Reference Design

Figure A-1 shows an example hardware design based on the FSM100xx ASIC for a sub-6 GHz Split (O-RU₆) and Integrated picocell. In both cases, this design implements the SCF FAPI protocol over an internal PCIe interface.

In the split Option 6 picocell case, the NPU implements the nFAPI control and data plane protocol encapsulation/decapsulation functions to provide the transport interface from the picocell to the O-DU₆. See SCF 225.1.0 [11].

1
2 In the integrated picocell case, the NPU implements higher layer protocol functions (L3 and
3 portions or all of L2.)
4
5

6

7

8

Annex 2:

This document references the following specifications from the Small Cell Forum available at https://scf.io/en/documents/all_documents.php

SCF 222.10.03: 5G FAPI PHY API Specification

SCF 082.09.05: nFAPI and FAPI Specifications

SCF 223.10.02: RF and Digital Front End Control API

SCF 224.10.01: 5G FAPI Network Monitor Mode API

SCF 225.2.0: 5G nFAPI Specifications.

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